

Taher Murtaza Rangwala

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EDUCATION

LJ Institute of Engineering & Technology (LIET), Ahmedabad, India

Expected April 2028

B.E. in Artificial Intelligence & Machine Learning — 3rd Year

- Relevant coursework: Machine Learning, Deep Learning, Data Structures & Algorithms, Full Stack Web Development

Holy Angels Convent School, Bharuch, India

[2023-2024]

Higher Secondary Education

PROJECTS

NyayaSetu (FinRAG) — Legal Conflict-Detection RAG System [HuggingFace Space](#)

- Built a two-stage hybrid retrieval pipeline (Pinecone BM25 sparse + Cohere dense embeddings, cross-encoder reranker) for high-precision legal judgment retrieval
- FastAPI backend with Groq LLaMA-3.3-70B for generation and Upstash Redis as the document store; React/Tailwind frontend
- Diagnosed a production out-of-memory crash to the BM25 encoder's pretrained corpus, and fixed it by training on a domain-specific corpus and lazy-loading components — cut encoder memory footprint from ~400MB to ~20MB

ERP for Flavour Compounding & Inventory Management — *manufacturing client*

- Built a locally-run ERP (Flask, React/Vite, Supabase) spanning inventory, formulation, and invoicing for a flavour-compounding business, packaged as a one-click desktop app
- Implemented AI-powered stock entry using Groq Llama-4 Scout for OCR on handwritten warehouse logs, paired with RapidFuzz matching to resolve product name variants against the catalog
- Built a live formulation costing engine computing batch cost price and profit margin in real time, plus reactive stock control blocking production on insufficient raw materials

Studyloop — AI-Powered Study Planner for JEE/NEET Aspirants [studiloop.me](#)

- Designed a Flask (sync CRUD) + FastAPI (async worker) split — Flask returns immediately while a Redis-queued worker handles slow Gemini calls for roadmap generation and answer grading
- Implemented SM2 spaced-repetition scheduling across a 17-table PostgreSQL schema (SQLAlchemy 2.0, Alembic migrations); Google OAuth + JWT auth
- Deployed on an Azure VM (Standard B2ats v2); React (Vite) frontend with live CI/CD with github actions.

InsightsDSA — AI-Powered DSA Tracker [insightsdsa.app](#)

- Built a LangGraph-orchestrated AI tutor with PostgreSQL-backed conversation checkpointing, supporting multi-provider LLMs (Gemini, OpenAI, Grok) via encrypted BYOK API keys (Fernet)
- Implemented an async background worker (Redis queue) that scores student understanding via Pydantic-structured LLM output after every session

FinAuto Sales Assistant — RAG Chatbot (*in development*)

- Building a narrow-scope RAG chatbot to handle pre-sales product Q&A for a Schedule III financial-statement SaaS platform, with automatic escalation to human demo booking for unresolved queries
- Implemented prompt-level and input-validation guardrails to block prompt injection and off-topic responses
- Integrating LLM observability (LangSmith) to trace every query end-to-end for debugging and quality monitoring

TECHNICAL SKILLS

Languages: Python, JavaScript, SQL

AI & Machine Learning: RAG architecture, Transformers, LangChain, Hybrid Search (BM25 + Dense), Cross-Encoder Reranking, Prompt Engineering, Scikit-learn, Pandas, NumPy

Backend & Data: FastAPI, Flask, REST APIs, PostgreSQL, Redis, SQLAlchemy, ChromaDB, Pinecone

Cloud & Tools: Microsoft Azure (VM deployment), Git, Linux fundamentals, React

Data Structures & Algorithms: Arrays, Hashing, Recursion, Two Pointers, Sliding Window, Sorting (Striver's A2Z)